

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on building systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, understanding natural language, and decision-making. AI technologies are widely used in modern applications such as chatbots, recommendation systems, self-driving cars, healthcare systems, and virtual assistants. Machine Learning (ML) is a subset of AI where systems learn patterns from data and improve over time without being explicitly programmed. Deep Learning is another important area of AI that uses artificial neural networks to process large amounts of data. Deep Learning powers technologies such as image recognition, speech recognition, and language translation. There are three main types of AI: 1. Narrow AI – designed for specific tasks. 2. General AI – capable of performing any intellectual task a human can do. 3. Super AI – a theoretical form of AI that surpasses human intelligence. Applications of AI include: - Healthcare diagnosis - Fraud detection - Autonomous vehicles - Smart assistants - Cybersecurity - Education systems Advantages of AI: - Automation of repetitive tasks - Faster decision-making - Improved accuracy - 24/7 availability Challenges of AI: - Ethical concerns - Bias in data - Privacy issues - High computational cost AI continues to evolve rapidly and is expected to play a major role in the future of technology and society.

Field	Example
Healthcare	Disease Detection
Finance	Fraud Detection
Transportation	Self-driving Cars
Education	Smart Learning Systems